

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2019

Subject: 392152 Chemistry II

Section: 15-18

Date: 22 January 2020

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 parts. There will be 9 pages (including this page) and they are worth 60 points which is 40% from total score.

- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are **NOT** allowed in the examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Relative atomic mass of H =1, He =4, C =12, N =14, O =16, F =19, Na =23, S =32, Cl =35.5,
Ar =40, K =39, Cu = 64, Zn = 65.5, Ag = 108, P = 30

Solvent	Boiling Point ($^{\circ}\text{C}$)	K_b ($^{\circ}\text{C}/\text{m}$)	Freezing Point ($^{\circ}\text{C}$)	K_f ($^{\circ}\text{C}/\text{m}$)
Water	100	0.512	0	1.86
Benzene	80.1	2.53	5.5	5.12
Acetic acid	118.1	3.07	16.6	3.90
Nitrobenzene	210.9	5.24	5.7	7.00
Phenol	182	3.56	43	7.40
Camphor	207.4	5.61	178.4	40.0

	Part 1 Multiple Choice	Part 2.1	Part 2.2	Total
score				

Part 1. Multiple Choice (15 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
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2				
3				
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15				

1. The phrases "particles close together and held in fixed positions" and "completely fills the container" apply, respectively, to
 - a. liquids and solids.
 - b. solids and gases.
 - c. gases and liquids.
 - d. liquids and gases.
2. "The pressure of a fixed amount of gas at a constant temperature is inversely proportional to the volume of the gas" is a statement of:
 - a. Charles' law
 - b. Boyle's law
 - c. Avogadro's law
 - d. Gay-Lussac's law
3. Indicate what the missing words are in the following statement of Charles' law: At constant pressure, the volume of a gas sample is _____ proportional to its _____ temperature.
 - a. directly; Celsius
 - b. directly; Kelvin
 - c. inversely; Celsius
 - d. inversely; Kelvin
4. Read the following sentences carefully.
 - (1) Sublimation and deposition are "opposite" changes of state.
 - (2) A pressure of 1 atmosphere is greater in magnitude than a pressure of 720 torr.
 - (3) The boiling point of a liquid can be increased by heating the liquid to a higher temperature.
 - a. All three statements are true.
 - b. Two of the three statements are true.
 - c. Only one of the statements is true.
 - d. None of the statements is true.
5. A *volatile* liquid would
 - a. have weak attractive forces between molecules.
 - b. evaporate "slowly" at room temperature.
 - c. have a low vapor pressure at room temperature.
 - d. have high boiling point.

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6. Compounds with relatively high vapor pressure have:
- high boiling points and weak intermolecular forces
 - low boiling points and weak intermolecular forces
 - high boiling points and strong intermolecular forces
 - low boiling points and strong intermolecular forces
7. A student puts 0.04 moles of lithium fluoride into 800 g of water. Determine the molality of the solution.
- 0.02 m
 - 0.05 m
 - 0.5 m
 - 2.00×10^4 m
8. Which one is not included in a colligative property?
- boiling point elevation
 - freezing point depression
 - vapor pressure elevation
 - osmotic pressure
9. What is the one of the main characteristics for all liquids?
- They are clear.
 - They are viscous.
 - They can be poured.
 - They are drinkable.
10. If helium and xenon behave ideally real gases, what is true about them?
- Average kinetic energy of helium is higher than that of xenon at the same temperature.
 - Helium rarely moves faster than xenon.
 - Both helium and xenon collide inelastically.
 - Effusion rate fraction of these gases is related to their densities.
11. Which statement is correct if a boy takes a helium balloon from a warm room into a cold room?
- The balloon remains the same.
 - The balloon gets bigger after walking into the cold room.
 - A hole can be seen.
 - The balloon seems to lose some helium gas but it shrinks.

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12. Which is an effect resulting in the faster vaporization rate of a liquid? *except*
- a. Increase in surface area.
 - b. Increase in temperature.
 - c. Increase in intermolecular force.
 - d. Decrease in viscosity.
13. Which statement is false?
- a. Carbon dioxide gas is a solute for soda.
 - b. Steel is one of the solid solutions.
 - c. Platinum cannot be a solvent for hydrogen gas.
 - d. Salt is a solute for sea water.
14. What are characteristics of colloid?
- a. Colloid solute is 10^{-4} cm.
 - b. It does not have a positive effect for Tyndall effect.
 - c. Sedimentation can be seen.
 - d. Its solute can be observed.
15. Which sweet tea would you expect to taste the sweetest?
- a. 1 M
 - b. 3.1 M
 - c. 3 M
 - d. 2.5 M

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Part 2 Write your answer in the space provided for each question.

Instruction: READ carefully

Students must provide calculation detail if CALCULATION is noted for any questions. Getting a full score cannot be provided if only answer without any detail is written.

1. Identify if the following (a-h) are properties of a solid, liquid or gas by putting each letter into the right column (3 points)

Properties	State of matter
a. Tightly packed in a regular pattern	Solid
b. move freely at high speeds	
c. Vibrate but do not move from place to place	
d. Free to flow to a shape of a container	Liquid.....
e. Close together with no regular arrangement	Gas.....
f. Well separated with no regular arrangement	
g. No defined volume or shape Intermolecular forces are strongest	

2. The pressure on 2.50 L of gas changes from 100 kPa to 40 kPa. What is the new volume if the temperature is unchanged? (CALCULATION, 2 points)

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3. If 10 liters of a gas at 37 °C are heated to 67 °C, what will be the new volume of this gas? (CALCULATION, 3 points)

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4. One mole of an ideal gas is at a pressure of 3,800 torr and a temperature of 25 °C. If the gas is heated at a constant volume until the pressure triples, what is the final temperature in degree Celsius (°C)? (CALCULATION, 3 points)

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5. What is the density of 1.00 mole of CO₂ at STP? (CALCULATION, 3 points)

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6. What is the relative rate of diffusion of SO₂ compared to He? Does SO₂ effuse faster or slower than He? (CALCULATION, 3 points)

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7. How many types of solid can be classified? Provide any differences those. (3 points)

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